

10.4 - Amortization Formula Explanations and Examples

Problem Solving, Math 1000

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1 Key Idea: Amortization Formula

In the US, almost all loans are a type of loan known as an *installment loan*, and almost all of these installment loans are based on monthly payments. Generally, the interest is compounded on a monthly basis as well. In such cases, the following formula, known as the *amortization formula*, gives the monthly payment M on a loan with principle P and APR r (as a decimal) which is to be paid out over T monthly installments:

$$M = P \frac{p(1+p)^T}{(1+p)^T - 1}, \text{ where } p = \frac{r}{12} \text{ is the monthly interest rate (as a decimal).}$$

1.1 Understanding Where the Amortization Formula Comes From

To understand this formula, we need to know a little bit about something called a *geometric sum*. A geometric sum can be explained most easily by using what mathematicians call "sigma notation" (which is just a lazy way of writing the sum of a bunch of different things). Sigma notation uses (as you might guess), the Greek letter sigma – more particularly, an upper case sigma, which looks like this: $\sum$.

To explain sigma notation, it might be easiest to start with some examples of the types of sums you can write with sigma notation. For instance, if I wrote

$$\sum_{n=1}^5 n$$

this represents the following sum

$$1 + 2 + 3 + 4 + 5.$$

And if I wrote

$$\sum_{n=1}^{100} n$$

it would represent the sum

$$1 + 2 + 3 + 4 + 5 + \dots + 97 + 98 + 99 + 100.$$

That is,

$$\sum_{n=1}^5 n = 1 + 2 + 3 + 4 + 5$$

and

$$\sum_{n=1}^{100} n = 1 + 2 + 3 + 4 + 5 + \dots + 97 + 98 + 99 + 100.$$

Similarly, if I wrote

$$\sum_{n=2}^5 \frac{3n+1}{2n}$$

I would be writing the sum

$$\frac{3(2)+1}{2(2)} + \frac{3(3)+1}{2(3)} + \frac{3(4)+1}{2(4)} + \frac{3(5)+1}{2(5)}$$

Are you starting to see how this notation works? If not, go look back at the examples given above a few more times until you start to see the pattern. Essentially, sigma notation can be broken down as follows:

1. the " $\sum$ " means "add all these things up",
2. the formula to the right of the $\sum$ symbol tells you what you're adding up (where the n is basically just a counter that goes up by one every time)
3. the part under the $\sum$ symbol (for instance, $n = 1$ in the first example, or $n = 2$ in the last example) tells you where to start your counter
4. the part above the $\sum$ symbol (for instance, 5 in the first example, or 100 in the second example) tells you where to end your counter.

One reason that sigma notation is useful is because it allows us to represent sums with a large number of terms without having to write all of those terms (and without having to write lots of ... or something like that). However, saving ink isn't the only reason we might care about sigma notation. More particularly, we know that certain "types" of sums can be computed by some pretty nice formulas. For example, I can use a formula for a type of sum called a *geometric sum* to calculate that

$$\sum_{n=0}^{1000} \frac{1}{7} \left(\frac{3}{19} \right)^n = 0.1696\overline{428571}.$$

Clearly, adding up all 1001 terms of this sum by hand would take a long time, so being able to compute this directly definitely is useful in some situations. We'll get to the formula that let me calculate the sum above in a minute, but we first need to define what a geometric sum is.

Definition 1 (Geometric Sum) Let r be a number, let a be a number, and let k be a positive whole number. Then, a sum that can be written in the form

$$\sum_{n=0}^{k-1} ar^n$$

is called a geometric sum.

The main idea behind geometric sums is that the ratio between successive terms is constant. For instance, consider the sum

$$1 + 2 + 4 + 8 + 16 + 32$$

Then, notice that 2, the second term of the sum, is just 2 times as much as 1, which is the first term of the sum. Similarly, 4, the third term of the sum, is just twice as much as 2, the second term of the sum. We can keep going with this, but you probably get the picture by now: If you take a term in the sum and divide it by the previous term, then the sum is geometric if you always get the same number.

Geometric sums are very nice for a couple of reasons, but the biggest reason by far is that they have a nice formula to compute their value. If r is a number and isn't equal to 1, then this formula can be stated as follows:

$$\sum_{n=0}^{k-1} ar^n = \frac{a(1 - r^k)}{1 - r}.$$

Of course, you might ask, what if our sum doesn't start with the counter n equal to zero? That is, what about sums like

$$\sum_{n=1}^{k-1} ar^n.$$

Well, these sums are pretty easy to deal with too – all you need to do is use the formula above and get rid of the first term of the sum (which is just equal to ar^0 , which simplifies down to a). That is, all you need to do is note that

$$\sum_{n=1}^{k-1} ar^n = \left(\sum_{n=0}^{k-1} ar^n \right) - a.$$

So, the formula for these types of sums is just

$$\sum_{n=1}^{k-1} ar^n = \frac{a(1 - r^k)}{1 - r} - a = \frac{a(1 - r^k) - a(1 - r)}{1 - r}.$$

Now, using this, we can derive the amortization formula. More particularly, suppose we have a loan and we make monthly payments of $\$M$ for T months on that loan at a monthly interest rate of p (written as a decimal). Then, the person we're paying the loan to doesn't get a lot of the money until a while into the future, which is inconvenient to them. So, many banks or other financial institutions which issue loans set up the monthly payments so that the present value of all the payments is equal to the amount owed.

That is, if the future value of the n^{th} month's payment you'll make is $\$M$ (since you pay $\$M$ every month), then that payment has a present value which we can find as follows:

$$\begin{aligned}\$M &= \$P(1+p)^n \\ \$P &= \frac{\$M}{(1+p)^n}.\end{aligned}$$

So, the total present value of the loan they made you is just

$$\frac{\$M}{(1+p)} + \frac{\$M}{(1+p)^2} + \frac{\$M}{(1+p)^3} + \dots + \frac{\$M}{(1+p)^{T-1}} + \frac{\$M}{(1+p)^T}.$$

But now, notice that:

$$\frac{\$M}{(1+p)^2} \div \frac{\$M}{(1+p)} = \frac{1}{1+p}$$

and that

$$\frac{\$M}{(1+p)^3} \div \frac{\$M}{(1+p)^2} = \frac{1}{1+p}$$

and, in fact, this holds for every term of the sum! So, this a *geometric sum*. More specifically, the present value of the loan is just

$$\sum_{n=1}^T \frac{\$M}{(1+p)^n} = \frac{\$M}{(1+p)} + \frac{\$M}{(1+p)^2} + \frac{\$M}{(1+p)^3} + \dots + \frac{\$M}{(1+p)^{T-1}} + \frac{\$M}{(1+p)^T}.$$

So, if the present value of the loan is set equal to the principle of the loan, $\$P$, then we can use the geometric sum formula to write

$$\$P = \sum_{n=1}^T \frac{\$M}{(1+p)^n}$$

But now, notice that we can solve for $\$M$ as follows:

$$\begin{aligned}\$P &= \sum_{n=1}^T \frac{\$M}{(1+p)^n} \\ \$P &= \sum_{n=1}^{(T+1)-1} \frac{\$M}{(1+p)^n} \\ \$P &= \frac{\$M \left(1 - \frac{1}{1+p}^{T+1}\right) - \$M \left(1 - \frac{1}{1+p}\right)}{1 - \frac{1}{1+p}}\end{aligned}$$

And, if we do a lot of algebra, this ends up simplifying to the following familiar formula when we solve for $\$M$:

$$\$M = \$P \frac{p(1+p)^T}{(1+p)^T - 1}.$$

So, there it is. The amortization formula is just what you get when you solve for M after setting the principle equal to the present value of all future payments (which you evaluate by noticing that its a geometric series)!

Of course, if all of this explanation is confusing, then ignore it. The main thing that matters is that you can *use* the amortization formula. Let's do an example to see how this works.

1.2 Example

Let's say you're buying a new car. Suppose you found a great car and negotiate a good price with the dealer: \$23,995 including taxes and license fees. In addition, suppose as well that the car manufacturer is offering a promotion where you can choose between a \$2,000 rebate or special financing at 1.2% APR for 60 months. If you choose the manufacturer's rebate, then you have to finance the balance at the dealer's rate of 6.48% APR for a 60-month loan. So, which one do you choose?

If you take the \$2,000 manufacturer's rebate and finance the balance through the dealer, you will have a principal of $P = 21995$, an APR of $r = 0.0648$, a monthly interest rate of $p = \frac{0.0648}{12} = 0.0054$, and a total of $T = 60$ monthly payments. Plugging this into the amortization formula gives monthly payments of:

$$M = (21995) \frac{(0.0054)(1.0054)^{60}}{(1.0054)^{60} - 1} = \$430.15$$

If you choose the special financing offer instead, you then have $P = \$23,995$, an APR $r = 0.012$, a monthly interest rate of $p = \frac{0.012}{12} = 0.001$, and a total of $T = 60$ monthly payments. The amortization formula gives monthly payments of

$$M = (23995) \frac{(0.001)(1.001)^{60}}{(1.001)^{60} - 1} = \$412.23$$

So, at the end of the day, you can see that the special financing offer is definitely better than the manufacturer's rebate. Over the life of the loan, you'd save over \$1,000 by choosing this option.

1.3 Amortization Identities

Often, when we're using the Amortization formula, we might want to figure out something other than the monthly payment. For example, we may know the monthly payment for a car advertised on television, but the principle, P , might not be given in the advertisement. Or, perhaps we know that we can pay off a certain amount of a debt every month (like, for example, a credit card balance) and we want to know how long it will take us to pay off the entire balance. Or, perhaps we've won the lottery, and we want to know whether we should choose the lump sum or the monthly payment as our payout option. Regardless, there are many examples of situations where we are dealing with some form of amortization and want to know something other than the monthly payment (or payout, in the lottery example).

To deal with these situations, knowing formulas to solve for M , P , and T will be very useful (the formula for p isn't particularly easy, so we'll skip it for now). These formulas can be stated as follows:

$$M = P \frac{p(1+p)^T}{(1+p)^T - 1}$$

$$P = M \frac{(1+p)^T - 1}{p(1+p)^T}$$

$$T = \frac{\ln\left(\frac{M}{M - pP}\right)}{\ln(1+p)}$$

In the last formula (the one where we solve for T), there is one caveat. Namely, if you end up trying to take the natural logarithm of a negative number when using this formula, then it means there will never be a large enough T to pay off the balance of the loan (or, if not a loan, then whatever the appropriate verbiage would be in that situation). That is, we can't take the natural log of a negative number and get a real number (its kind of like taking the square root of -1 in that respect), so there isn't any real solution – which means (in the loan case, for example) that the payment is too small to keep up with the interest, and the balance will *never* be paid off!

We'll do a lot of examples using these formulas in lecture, so hopefully they'll become familiar soon. For now, just reference this sheet when you need to solve these types of problems, since all of this really just boils down to some calculator work at the end of the day.